Low-Level Design (LLD)

Legal Document RAG Assistant

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1 Module-Level Design

1.1 Document Processing Module

Loads and preprocesses PDF documents.

1.2 Chunking Module

Splits text into overlapping chunks.

1.3 Embedding Module

Generates embeddings using MiniLM.

1.4 Vector Storage Module

Stores embeddings in ChromaDB.

1.5 Retrieval Module

Fetches relevant chunks using similarity search.

1.6 Query Processing Module

Formats query and retrieved context into a prompt.

1.7 Graph Execution Module

Executes LangGraph workflow.

1.8 HITL Module

Handles escalation logic.

2 Data Structures

2.1 Document Representation

```
{
  content: string,
  metadata: { page_number }
}
```

2.2 Chunk Format

```
{
  chunk_text: string,
  source: document_id
}
```

2.3 Embedding Structure

```
vector: [float values]
```

2.4 Query-Response Schema

```
{
  query: string,
  answer: string
}
```

2.5 State Object (LangGraph)

```
{
  query: string,
  route: string,
  answer: string
}
```

3 Workflow Design (LangGraph)

3.1 Nodes

- route_node
- process_node
- hitl_node

3.2 Edges

- route → process
- route → hitl

3.3 State Flow

Query → Route Decision → Processing / Escalation

4 Conditional Routing Logic

4.1 Answer Generation Criteria

- Relevant chunks found
- Query is simple

4.2 Escalation Criteria

- Low confidence
- No relevant context
- Complex legal query

5 HITL Design

5.1 When Triggered

- Missing context
- Ambiguous question

5.2 Process

Query → Human Expert → Response Returned

6 API / Interface Design

6.1 Input

User query (string)

6.2 Output

Answer (string)

6.3 Flow

User → CLI → Graph → Response

7 Error Handling

7.1 Missing Data

Return: *"Not found in document"*

7.2 No Relevant Chunks

Fallback to HITL.

7.3 LLM Failure

Retry or escalate.